Could we simplify the picture, and make it closer to known territories, by saying the first subsets of the context is that of propositions/expressions, and that it is a full subset (the whole context is made out of expressions/propositions). If we go with the proposition we lean towards the logic presentation of context, if we use expressions we lean towards a Spinoza/Deleuze view of expressions. The former enables us to make Borrow-Work-Replay intellegible at this pre-historical level. The former is what enables us to describe the given context as perspectivist. The next level is then between 'real', where here we look to Lautman and his movement 'metaphysics problem -> effective mathematics -> physical theories of the real', and 'abstract' (what could be a better word? or a good anchor/ground|author/concept?). Then next give us agents, where here we use Simondon and Kauffman to distinguish those propositions/expressions that have entered another realm, the non-ergodic, the individualizing, etc. And eventually we have complex language (where maybe Manin & Marcolli could be a good anchor, or Peirce's semiotics?).
The pre-philosophical could then be restricted to the "contravariant" movement, "la remontée vers le contexte" as I said above, both the necessity and capacity of understanding for those communicating agents and their games/lives. And we only notice that in so doing they could help the infinite in its "quest to be itself", as a side remark. Just like creating a new context, renewing the possibility of a cascade of subsets that reach the necessity/possibility of understanding and encoding a context is a byproduct of a community busy understanding its own context.

Does that "clean up" the first version?
Any remarks about this new version?

I'll check the proposed anchors, especially whether propositions and expressions can serve as two readings of one starting point, and where the "real/abstract" distinction would need a different formulation.

Yes—this version is substantially clearer, especially because understanding no longer has to fulfil a cosmic obligation. Situated beings pursue their lives and inquiries; their achievements can become conditions for further activity. Renewal becomes a possible consequence of that process, not a demand that the infinite imposes on it.

I read your second "former" as "latter": propositions make the logical reading available; expressions support the Spinoza–Deleuze reading.

I would keep this direction, with a few distinctions.

1. Make "expression" the broad starting point, and propositions a particular articulation

Saying that the context is expressive throughout is a coherent philosophical starting point. But saying that everything is a proposition is a considerably stronger—and different—claim.

In the Spinoza–Deleuze register, expression is not necessarily a statement, representation, or message addressed to someone. Substance expresses itself through attributes and modes; an existing being need not describe anything to participate in this expressive relation. Deleuze, Spinoza et le problème de l'expression.

A proposition, by contrast, belongs to an organization of claims, judgments, consequences, and justification. An organism and a proposition about that organism cannot simply be exchanged.

I would therefore formulate your starting point as:

> We consider a context through its expressions and their relations. Logical presentations articulate some of those relations as propositions, constructions, and rules of transformation.

This preserves the possibility of the stronger metaphysical reading—everything participates in expression—without making everything already documentary content.

There is a particularly useful distinction here:

- Being an expression of a context.
- Producing expressions through which that context can be understood.

The second can emerge within the first. That is almost exactly the turn your cascade needs: some beings become capable of interpreting, articulating, and communicating aspects of the context they already belong to.

Also, if the first "subset" is the whole context, it is really the starting register, not yet a subdivision. Calling it an expressive reading of context would be clearer than "full subset."

2. The logical reading makes BWR legible, but does not supply it by itself

Propositions alone do not yet give borrowing, work, or replay. Their relations and operations do.

In a logical presentation, one can recognize:

- borrowing as using assumptions, resources, or previously established constructions;
- work as making a derivation or construction;
- return as articulating what has been achieved and under which conditions;
- replay as applying that achievement in another admissible situation.

That is enough to make BWR intelligible at the level of your orientation. It does not require claiming that every physical process is literally a proof, or that logical applicability automatically produces actual uptake.

Likewise, expression alone does not establish perspectivism. The further ingredient is differentiated expression: the context becomes determinate differently through different relations and organizations. Your opening can state this explicitly rather than derive it from infinity.

The two readings then cooperate without being synonyms: expression concerns how a context becomes manifest through its beings; logic concerns disciplined articulation and transformation.

3. Replace "real/abstract" with "concrete existence/ideal structure"

That would be my preferred wording for this particular picture. "Real/abstract" misleadingly suggests that mathematics or general forms are less real.

You already have an anchor for ideal structure in Lautman. I would not add another philosopher merely to fill that branch. His problematic relations, mathematical realizations, and relation to physical theories give you ways of discussing different forms of determination—not a simple division between reality and unreality.

But these should be related aspects or registers, rather than entirely separate populations. A physical configuration can realize a mathematical structure; a written proof is a concrete occurrence carrying an ideal construction. The relation between the two is part of what needs explaining.

Peirce offers a useful, more specific companion: repeatable forms and their particular instances. His type/token distinction separates a significant form from an occurrence embodying it. That is especially pertinent once the cascade reaches documentation and reuse. It is not a complete ontology of physical versus mathematical being. Peirce, Prolegomena to an Apology for Pragmaticism.

I would therefore use:

> Concrete processes and beings; ideal structures through which their relations can be articulated.

And keep Lautman's movement as a relation among problematics, mathematics, and physical theorization—not as a chronological manufacturing chain through which metaphysics produces matter.

4. Agency needs a positive distinction, not just entry into non-ergodicity

Simondon and Kauffman remain appropriate, but "non-ergodic" cannot do all the classificatory work. Failure to explore a space of possibilities does not, by itself, make something an agent.

The stronger step is:

> Some organizations participate in maintaining—and sometimes transforming—the conditions of their own continuation.

That gives your "fuzzy game" its substance: what happens can support or undermine the organization's continued activity. Montévil–Mossio's mutually maintained constraints provide a precise biological candidate for this distinction. Biological organisation as closure of constraints.

Simondon contributes individuation with a milieu; the biological closure accounts contribute particular forms of self-maintenance. These are connected resources, not interchangeable definitions. Historically contingent possibilities and non-ergodicity can then explain aspects of their development without becoming the sole boundary between life and non-life.

5. Peirce first for the language transition; Manin–Marcolli for a mathematical investigation

For the orientation itself, Peirce is the stronger primary anchor. His sign–object–interpretant relation lets you describe how something becomes significant through interpretation, and how an interpretation can support further interpretation. It also prevents meaning from being reduced to a direct pairing between a symbol and a thing. Peirce's definitions of the sign.

Complex language would then involve particular organizations of semiosis: reusable signs, compositional expression, inference, communication, and collective revision. Peirce's semiosis is broader than human language, so this remains a further development rather than its definition.

Manin and Marcolli fit as a more specialized mathematical resource. Their Semantic Spaces explicitly addresses the relationship between textual resources, language in human minds, and communication, using geometric models of semantics. That is relevant to the community/documentation distinction, but is not itself an account of why linguistic agents emerge. Manin–Marcolli, Semantic Spaces.

So I would give them different jobs: Peirce for the passage into interpretation and inquiry; Manin–Marcolli for candidate structures describing semantic organization.

6. The “remontée” is now the most compelling part

Your revision removes the need to claim that the universe was destined to manufacture its own cartographers.

The remaining idea is strong:

> Some beings whose activity depends on their context develop ways of understanding it. Through communication, those understandings become reusable beyond the situation that produced them. Their reuse can change the conditions under which subsequent beings act and understand.

There are two qualifications worth retaining. First, needing an adequate orientation does not mean needing exhaustive knowledge: understanding is costly, selective, and situated. Second, new contexts are a possible achievement, not an automatic byproduct. Documentation can also reinforce a closed interpretation or become unusable.

“Contravariant” can remain an evocative name for this return toward the conditions of one’s existence, provided it is not yet presented as categorical contravariance.

I would not quite restrict all pre-philosophy to the ascent: the initial decision to understand being through expression is already philosophical orientation. But you can certainly concentrate the speculative claim there, leaving the infinite’s “quest to be itself” as an optional interpretation rather than a premise.

The resulting picture is:

> An expressive context supports concrete organizations; some sustain their own activity, some interpret and communicate, and some organize those interpretations into reusable understanding. In doing so, they may transform the context of further activity without having set out to serve a cosmic purpose.

That cleans up the first version while preserving its ambition. Its crucial distinction is now not finite versus infinite, but participating in a context versus articulating it—and the possibility that the second activity changes the conditions of the first.

Q

Can we now rethink the relation between a) the "covariant" pre-historical & "contravariant" pre-philosophical, and b) the series of separations and there echoing integrations ?

Yes. I think we can now relate them more clearly: the two movements describe where the process goes; separation and integration describe how it proceeds without destroying what makes it possible. They are two intersecting distinctions, rather than two names for the same distinction.

1. The downward movement produces beings capable of a return movement

Your “covariant” pre-history follows how a context supports determinate organizations, some of which sustain themselves, interpret their surroundings, and communicate. Your “contravariant” pre-philosophy follows how those organizations articulate the conditions, relations, and problems through which they exist.

The important connection is:

> A context supports beings capable of articulating it; their articulations can become conditions for further beings and activities.

The return is therefore not simply tracing the outward route backwards. A community does not reconstruct the entire genesis of its existence. It produces a particular understanding, with particular powers of explanation and use. That understanding can introduce distinctions and possibilities that were not available to the community before.

This also lets us distinguish two achievements: understanding an enabling context and producing something that becomes enabling context. They can coincide, but neither guarantees the other.

2. Separation and integration operate in both directions

The tempting assignment would be “pre-history separates; pre-philosophy integrates.” It catches something, but misses half the process.

Movement	What separation contributes	What integration contributes
Context → situated beings and activities	Distinct organizations, capacities, interfaces, and ways of participating can arise.	Relations with a milieu make those distinctions viable and effective.
Situated activity → articulated context	Inquiry distinguishes concepts, dependencies, assumptions, and differences between perspectives.	Those distinctions become connected explanations, reusable constructions, and conditions for further inquiry.

An organism, for example, is not produced by separation alone: its relative independence depends on organized relations within itself and with its milieu. Conversely, understanding is not integration alone: a good explanation often succeeds by discovering a distinction that an earlier account obscured.

So the stronger formulation is:

> The outward movement makes differentiated participation possible; the return movement makes the relations of that participation articulable. Each requires both separation and integration.

This is where your mirror becomes reciprocal rather than merely symmetrical. What enables a distinction in one setting can become something explicitly distinguished in another; what has been distinguished can then support a new integration.

3. The mirror connects practical separations with their enabling conditions

This gives us a useful interpretation of the K/M investigation.

On K, we encounter a practical requirement: two responsibilities must not be confused. On the contextual side, we investigate the relations that allow them to remain distinct while working together.

For example:

- Participants must remain distinct from their documentary contributions. The corresponding integration question is how contributions can form shared understanding without requiring exhaustive representations of their contributors.
- Local work must remain situated. The integration question is under what compatibility conditions its results can contribute to something beyond their original situation.
- Stabilized concepts must remain distinguishable from their possibilities of variation. The integration question is how reusable forms can participate in families without erasing the differences relevant to further work.

The return movement can then disclose further separations. Trying to integrate local results may reveal that apparent agreement concealed different assumptions. Trying to reuse a concept may reveal the need to distinguish agreement at one instance from agreement under variation.

That is precisely the “secondary separations” idea: integration does not merely reconnect previously separated pieces; it can teach us where another distinction is needed. Reciprocally, making that distinction can enable an integration that was previously unavailable.

Neither K nor M exclusively owns separation or integration. The proposed correspondence concerns how their respective operations make one another intelligible and usable.

4. The resulting picture is recursive, without making recurrence compulsory

We can now read BWR through this arrangement:

> Borrow enabling relations; work through explicit distinctions; return an articulated achievement; replay it through another situated activity.

Successful replay may preserve an existing possibility, solve a problem, or help constitute a changed context. It need not always generate novelty. But when it does change the context, the next movement starts somewhere different: previous understanding has become part of what later activity encounters and uses.

Your pre-philosophical intuition then acquires a particularly good form:

> The context does not need to command its inhabitants to understand it. Some inhabitants’ activities make understanding consequential; their shared understanding can, in turn, alter the conditions of inhabiting it.

The infinite’s “quest to be itself” can remain an interpretation of this circulation, without carrying the explanatory burden.

One terminology boundary should remain explicit: these are currently generative and regressive directions, not established categorical variances. In particular, both functors in Allow ⊣ Require are ordinarily covariant; being a right adjoint does not mean being contravariant.

The relation I would retain is therefore: pre-history follows the emergence of situated capacities; pre-philosophy follows their articulation of context; the separation/integration mirror describes how each movement preserves distinctions while enabling relations—and sometimes discovers new distinctions through those relations.
Worked for 1m 12s • 5:28 AM

Q

This is a proposition by Claude, which I find quite interesting, since I agree "fullness" wasn't the right frame for propositions-in-context, and that it could give us the key to understand the structural influence of the two kinds of infinities in our story, and link the Spinoza-Deleuze track (conatus) with the logical track (Lawvere-Kock and its surroundings, or maybe something else ~ that which probably Deleuze was looking for in his study of dx/dy and the likes), that are, at this pre-level, the two main views we have.
What do you think?

Could you paste Claude's proposition? It didn't come through. The precise replacement for "fullness" matters here—especially for assessing how it connects the two infinities, conatus, and infinitesimal variation without treating them as interchangeable.
Worked for 9s • 5:41 AM

Where the revision is still in trouble

1. Fullness and BWR cannot both live at level 1. This is the sharpest remaining problem and neither of you drew it out. Codex gets close — "saying that everything is a proposition is a considerably stronger and different claim" — but stops before the consequence. An organism can be an expression of substance without stating anything, so "the whole context is expressions" is coherent. "The whole context is propositions" makes organisms into claims, so fullness is false on that reading. Therefore the fullness thesis forces the expression reading, and the expression reading is the one that does not give you Gentzen's borrow/work/return. You cannot have two readings of one starting point and fullness and BWR at the ground.

The repair is better than the problem, though. There are two BWRs, not one. On the proposition reading it's intro/elim harmony — assume, derive, discharge, apply. On the expression reading it's composition of powers in the Ethics — affect, be affected, persevere. And arc/21's one durable philosophical result was the conatus repair, which relocates striving from the infinite to the situated being. So the expressionist BWR is the conatus-BWR, and it is the version the file already converged on. Say there are two, name which level each lives at, and the tension dissolves into a finding.

I'll check the proposed repair against the logical and Spinozist meanings. The question is whether these are two realizations of BWR, or two distinct processes whose relationship BWR invites us to investigate.

Yes—there is a useful finding here. Claude separates the continuation of an activity from the reuse of an articulated achievement. That could help connect the expressionist and logical views without making the world itself a collection of propositions.

But I would call these two proposed readings of BWR, not yet two established instances. The distinction is convincing; their common structure remains something to discover.

1. The diagnosis is right, but the incompatibility is narrower

We cannot identify these two claims:

- Everything participates in the context's expression.
- Everything is a proposition in a logical presentation.

However, an expressive account of the context and logical presentations of it can coexist at the same level of discussion. The missing ingredient is their relationship, not necessarily an extra rung in the cascade.

A being can express its context without stating anything. A logical presentation can articulate aspects of that being without claiming to exhaust it. The boundary between expression and logical articulation is itself productive: it gives us something to investigate rather than requiring a choice between philosophy and logic.

So I would replace “fullness forces one reading” with:

> Expression supplies the broad orientation; logical articulation supplies particular, accountable ways of working within it.

2. Claude’s two BWRs reveal two different responsibilities

The proposed comparison is approximately this:

	Logical reading	**Expressionist reading**
Starting conditions	Assumptions, resources, established constructions	Relations and powers through which situated activity is possible
Work	Deriving or constructing	Acting and being affected
What can continue	An achievement through further application	An activity or organization through maintained or transformed conditions
Central question	What licenses this reuse?	What enables this continuation?

On the logical side, the implication example is precise: derive B under assumption A, discharge A to obtain A ⇒ B, then apply that result when A is supplied. Introduction/elimination harmony concerns whether those construction and use rules fit together. It does not by itself establish actual uptake, sustainable inquiry, or contextual renewal. Pfenning, Harmony.

On the expressionist side, conatus concerns a thing’s striving to persist, which Spinoza identifies with its actual essence. But “affect, be affected, persevere” is not already a borrow–work–return–replay cycle. It does not automatically specify a return, a transferable achievement, or replenishment of the conditions supporting others. Moreover, Spinoza attributes conatus to things generally, not only organisms or communicating agents. Ethics, III, propositions 6–8.

The valuable connection is therefore a question:

> Under what conditions does the reuse of achievements contribute to the continuation—and possible transformation—of the activities that make achievements possible?

That is close to Hoapa’s central concern. A proof can remain reusable while the community capable of understanding it disappears. Conversely, a community can maintain itself while producing little that others can independently reuse. Neither responsibility subsumes the other.

One record correction: our current opening (workflow/openings.md:135) retains the conatus reading as an alternative to the cosmic-contract picture, not a settled replacement. It explicitly leaves sustainable BWR

underived from striving alone.

3. This gives the two infinities a role without assigning one to each BWR

I would not make “large infinity = expressionist BWR” and “infinitesimal = logical BWR.” Both readings encounter both concerns.

Within our proposed interpretation:

- The large pole asks how situated activity relates to enabling conditions that its own organization or account does not exhaust.
- The infinitesimal pole asks how a determinate organization can carry differences and possibilities not exhausted by its currently stabilized form.

Their meeting question becomes:

> How can something remain sufficiently determinate to act and be reused, while remaining sensitive to changes in itself and its relations?

That connects conatus with variation more carefully than identifying conatus with change. Perseverance sometimes requires resisting variation; sometimes it requires changing. Conversely, a possible variation need not support continuation.

This makes the infinitesimal interesting not because it is inherently creative or caring, but because it can help distinguish preserving a present configuration from preserving specified capacities to respond.

4. Kock–Lawvere could help articulate the connection—not certify it

Here there is a concrete mathematical prospect. A smooth interpretation can distinguish:

- an element in its present state;
- its infinitesimal variations;
- constraints relating those variations to other elements;
- whether those variations extend into admissible further activity.

Kock–Lawvere gives a precise language for the first-order part: a map from the basic infinitesimal object into the smooth line is determined by a value and a slope. But infinitesimal variation, a chosen dynamics, and successful continuation remain different structures. Kock, Synthetic Differential Geometry, §§I.1, I.7–8.

For Hoapa, the proposed bridge would be: logical documentation can carry guarantees about the variations and continuations relevant to the activity it describes. The activity supplies the stakes; the mathematics makes particular preservation claims expressible and checkable.

Deleuze’s interest in differential relations is relevant to this search: he uses them to investigate how relations disclose determinations not given by considering their terms separately. That is a philosophical resource, not evidence that Kock–Lawvere realizes his programme. Deleuze, lecture of 29 April 1980. In particular, the synthetic derivative is obtained through a unique coefficient—not by dividing by a nilpotent infinitesimal.

This also sharpens the mirror. We separate an activity’s conditions of continuation from the conditions under which its documented achievements may be reused. Then we investigate integrations through which each

can support the other, without confusing a valid inference with a viable practice.

So I would retain Claude's repair in this form:

> BWR is the proposed meeting ground between situated continuation and accountable reuse. Conatus helps articulate the first; logical construction and application help articulate the second. Their connection—and its capacity to sustain renewal—is a principal question of Hoapa, not something already supplied by their shared name.